Christopher Lam (he/him)

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EDUCATION

University of British Columbia, Vancouver BC Expected Graduation: May 2029

Bachelor of Applied Science – Integrated Engineering

- Relevant Courses: Engineering Design, Experimental Physics Lab, Electricity & Magnetism
- CGPA: 82% (A–, UBC percentage scale)

KEY COMPETENCIES

Electronics & Embedded: Analog circuits, LED drivers, RP2040, Arduino, e-ink displays, low-power design
Robotics & AI: LeRobot, SmolVLA (vision-language-action), SO-101 arm, teleoperation, MediaPipe
CAD & Fabrication: Inventor, Fusion 360, CNC milling (CAD to G-code), FDM 3D printing, soldering
Programming & Test: Python, Arduino C, JavaScript, HTML/CSS; oscilloscopes, signal generators
Design & Writing: Technical writing, typography, graphic design, video; Japanese (conversational)

PROJECTS

Voice-Controlled Robot Arm — 2nd of 180 | Physical AI Sprint Hackathon (Nebius & NVIDIA) Aug 2026

- Turned Meta Ray-Ban glasses into a live perception feed that re-prompts the VLA controller
- Programmed robot-side control; working demo in five hours with a team formed that morning

SO-101 Robot Arm — Autonomy and Control | Personal Project Jul – Aug 2026

- Ran it on SmolVLA (vision-language-action), so autonomy didn't depend on precise teleop
- Dropped MediaPipe hand-tracking teleop (no depth or pitch); mapped an Xbox controller
- Wrote a safety layer so the arm cannot fold into itself, drive into the table, or reach too high

Tally — One-Button Counter | Personal Project Jun – Aug 2026

- Wrote RP2040 firmware for count, lock, and reset from one button, shown on an e-ink display
- Tuned the click through four housings in one day; split the shell so assembly drove the design

Custom LED Desk Lamp | Personal Project 2025

- Balanced a spring-loaded arm to hold any position without locks across its full sweep
- Built an analog LED driver for a 3000K COB emitter, choosing analog over PWM to kill flicker

NOTA Magazine — Writer & Designer | Two published issues May – Nov 2024

- Reported and wrote two features, then set the type, graphics, and layout each one ran in

LABORATORY EXPERIENCE

Experimental Physics Laboratory (PHYS 159) | University of British Columbia Jan – Apr 2026

- Characterized microphone transducer response to calibrate the measurement chain
- Captured waveforms on oscilloscopes and signal generators; propagated uncertainty throughout

TEAMS, LEADERSHIP & WORK

UBC Solar — Strategy & Simulation Designer | Vancouver, BC Sep 2026 – Present

- Designing driver, pit, and strategy dashboards from the car's energy model to set race pace

Scouts BSA Troop 941 — Senior Patrol Leader & Eagle Scout | Pleasanton, CA 2018 – 2025

- Elected to lead a 100-member troop; Eagle Project: public bike repair station, 15+ volunteers

COVID-19 Emergency PPE Production | Pleasanton, CA Mar – Aug 2020

- Self-started; scaled to 3,500+ printed face shields by tuning print settings and assembly order

In-N-Out Burger — Store Associate | Pleasanton, CA 2022 – 2025

- Advanced to Level 4 within two years; trained 10+ new associates on standard work